

Index Notation, Kronecker Delta and Levi-Civita Symbol

Part 3 of the OT Science Formula Completion Workbook

OT Science Notes

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1 Index Notation, Kronecker Delta and Levi-Civita Symbol

Index notation compresses vector and tensor formulas into short expressions. It is especially useful when proving vector identities. The price is that every index must be handled carefully.

1.1 Einstein Summation Convention

When an index is repeated once as a subscript or repeated in a product in Euclidean notation, it is summed over its full range. In three-dimensional Cartesian coordinates, the range is usually 1, 2, 3.

Einstein Summation

$$a_i b_i = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

$$A_{ij} b_j = A_{i1} b_1 + A_{i2} b_2 + A_{i3} b_3.$$

$$A_{ij} B_{ij} = \sum_{i=1}^3 \sum_{j=1}^3 A_{ij} B_{ij}.$$

$$(AB)_{ij} = A_{ik} B_{kj}.$$

A free index appears on both sides of an equation and is not summed. A dummy index is repeated and summed. Dummy indices can be renamed as long as no conflict is created.

1.2 Kronecker Delta

The Kronecker delta is the identity tensor in component form. It replaces one index by another.

Kronecker Delta Identities

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

$$\delta_{ij} a_j = a_i, \quad \delta_{ij} a_i = a_j.$$

$$\delta_{ij} \delta_{jk} = \delta_{ik}, \quad \delta_{ii} = 3.$$

$$\delta_{ij} A_{ij} = A_{ii} = \text{tr}(A).$$

$$\partial_i x_j = \frac{\partial x_j}{\partial x_i} = \delta_{ij}.$$

$$\partial_i x_i = 3, \quad \partial_i r = \frac{x_i}{r}, \quad \partial_i (r^2) = 2x_i.$$

1.3 Levi-Civita Symbol

The Levi-Civita symbol tracks orientation and cross products. It is zero whenever any two indices are repeated.

Levi-Civita Identities

$$\varepsilon_{ijk} = \begin{cases} +1, & (i, j, k) \text{ is an even permutation of } (1, 2, 3), \\ -1, & (i, j, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0, & \text{two indices are repeated.} \end{cases}$$

$$(\mathbf{a} \times \mathbf{b})_i = \varepsilon_{ijk} a_j b_k.$$

$$\varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}.$$

$$\varepsilon_{ijk} \varepsilon_{ijm} = 2\delta_{km}, \quad \varepsilon_{ijk} \varepsilon_{ijk} = 6.$$

$$\varepsilon_{ijk} \varepsilon_{lmn} = \begin{vmatrix} \delta_{il} & \delta_{im} & \delta_{in} \\ \delta_{jl} & \delta_{jm} & \delta_{jn} \\ \delta_{kl} & \delta_{km} & \delta_{kn} \end{vmatrix}.$$

1.4 Vector Calculus in Index Notation

Vector Operations in Index Form

$$(\nabla f)_i = \partial_i f.$$

$$\nabla \cdot \mathbf{A} = \partial_i A_i.$$

$$(\nabla \times \mathbf{A})_i = \varepsilon_{ijk} \partial_j A_k.$$

$$\nabla^2 f = \partial_i \partial_i f.$$

$$(\nabla \mathbf{A})_{ij} = \partial_j A_i.$$

$$(\nabla \cdot \mathbf{T})_i = \partial_j T_{ij}.$$

1.5 Proof Pattern: Curl of Curl

The index method is powerful because the proof becomes a controlled substitution. Starting with

$$(\nabla \times (\nabla \times \mathbf{A}))_i = \varepsilon_{ijk} \partial_j (\nabla \times \mathbf{A})_k,$$

use

$$(\nabla \times \mathbf{A})_k = \varepsilon_{klm} \partial_l A_m.$$

Then

$$(\nabla \times (\nabla \times \mathbf{A}))_i = \varepsilon_{ijk} \varepsilon_{klm} \partial_j \partial_l A_m.$$

Using the contraction identity gives the familiar result

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

Index-Notation Mistakes

1. Do not repeat the same index more than twice in one term unless you have explicitly defined what it means.
2. Do not change a free index on one side of an equation without changing it on the other side.
3. Dummy indices may be renamed, but free indices may not.
4. The identity $\varepsilon_{ijk} \varepsilon_{ilm}$ depends on which index is contracted. Check the repeated index carefully.

Quick Drills and Practice: Index Notation

Level A: Expansion and simplification

1. Expand $a_i b_i$ in three dimensions.
2. Expand $A_{ij} b_j$ for $i = 1$.
3. Simplify $\delta_{ij} a_j$.
4. Simplify $\delta_{ij} A_{jk}$.
5. Simplify $\delta_{ij} \delta_{jk} \delta_{kl}$.
6. Simplify $\delta_{ij} A_{ij}$.
7. Compute δ_{ii} in three dimensions.
8. Compute $\partial_i x_j$.
9. Compute $\partial_i x_i$.
10. Compute $\partial_i (x_i x_j)$.

Level B: Levi-Civita

1. Evaluate ε_{123} , ε_{132} , ε_{221} , ε_{312} .
2. Expand $(\mathbf{a} \times \mathbf{b})_1 = \varepsilon_{1jk} a_j b_k$.
3. Show that $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ using ε_{ijk} .
4. Simplify $\varepsilon_{ijk} \varepsilon_{ijm}$.
5. Simplify $\varepsilon_{ijk} \varepsilon_{ijk}$.
6. Use $\varepsilon_{ijk} \varepsilon_{ilm}$ to simplify $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d})$.

Level C: Vector identities

1. Prove $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ using index notation.
2. Prove $\nabla \times (\nabla f) = \mathbf{0}$ using index notation.
3. Prove $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$.
4. Prove $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$.
5. Prove $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$.

Level D: Derivative identities

1. Show that $\partial_i r = x_i/r$, where $r = (x_j x_j)^{1/2}$.
2. Show that $\partial_i (1/r) = -x_i/r^3$.
3. Compute $\partial_i (x_i/r^3)$ for $r \neq 0$.

4. Compute $\partial_j(x_i x_j)$ in three dimensions.
5. If $T_{ij} = f\delta_{ij}$, compute $\partial_j T_{ij}$.